

**POSTS AND TELECOMMUNICATIONS INSTITUTE OF TECHNOLOGY**  
**COURSE: INT1313 DATABASES - SEMESTER 1, 2026-2027**  
**Phase 1 Progress Report - Group of 3 Students**  
**TOPIC 7: FOOD DELIVERY AGGREGATOR (MINI-DOORDASH)**

## **1. INTRODUCTION & PROBLEM STATEMENT**

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The objective of this project is to architect a robust relational database for a Food Delivery Aggregator ecosystem. The system is designed to seamlessly connect three primary entities: restaurants, customers, and delivery drivers. The database must maintain absolute data integrity while tracking the entire lifecycle of an order from initial placement and food preparation to real-time delivery status.

**Database Challenge:** The core complexity lies in managing complex multi-way joins across Customer, Restaurant, Driver, and Order tables. Additionally, the design must lay the foundation for handling geometric and location coordinates via spatial extensions for tracking driver and delivery hub locations.

## **2. SYSTEM REQUIREMENTS (ISO/IEC/IEEE 29148)**

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In accordance with the Software Requirements Specification (SRS) standards, the system scope is delineated into user expectations, system capabilities, and specific technical constraints.

### **2.1 User Requirements vs. System Capabilities**

- **User Requirements:** Customers need to browse menus and place orders securely. Restaurants require an interface to manage menu items and accept orders. Drivers need a system to receive dispatch requests and update their location.
- **System Capabilities:** The database must utilize Extended ER (EER) features (specialization/generalization) to manage a central User superclass that delegates attributes to Customer, Restaurant, and Driver subclasses. The system must also process multi-party transactions reliably.

### **2.2 Functional Requirements (Data-Centric)**

- The system must allow a Driver to update the status of an order (e.g., from 'PICKED\_UP' to 'DELIVERING').
- The system must compute the total order amount automatically by joining the OrderItem quantities with the corresponding MenuItem prices.
- The system must enforce constraints to prevent an Order from being deleted if a Driver account is deactivated (setting the foreign key to NULL instead of cascading deletion).

### **2.3 Non-Functional Requirements**

- **Performance:** Multi-way joins involving Orders, Users, and OrderItems must execute efficiently, returning aggregate data for dashboards in under 250ms.
- **Security:** Sensitive customer data, particularly passwords and preferred payment methods, must be encrypted at rest.

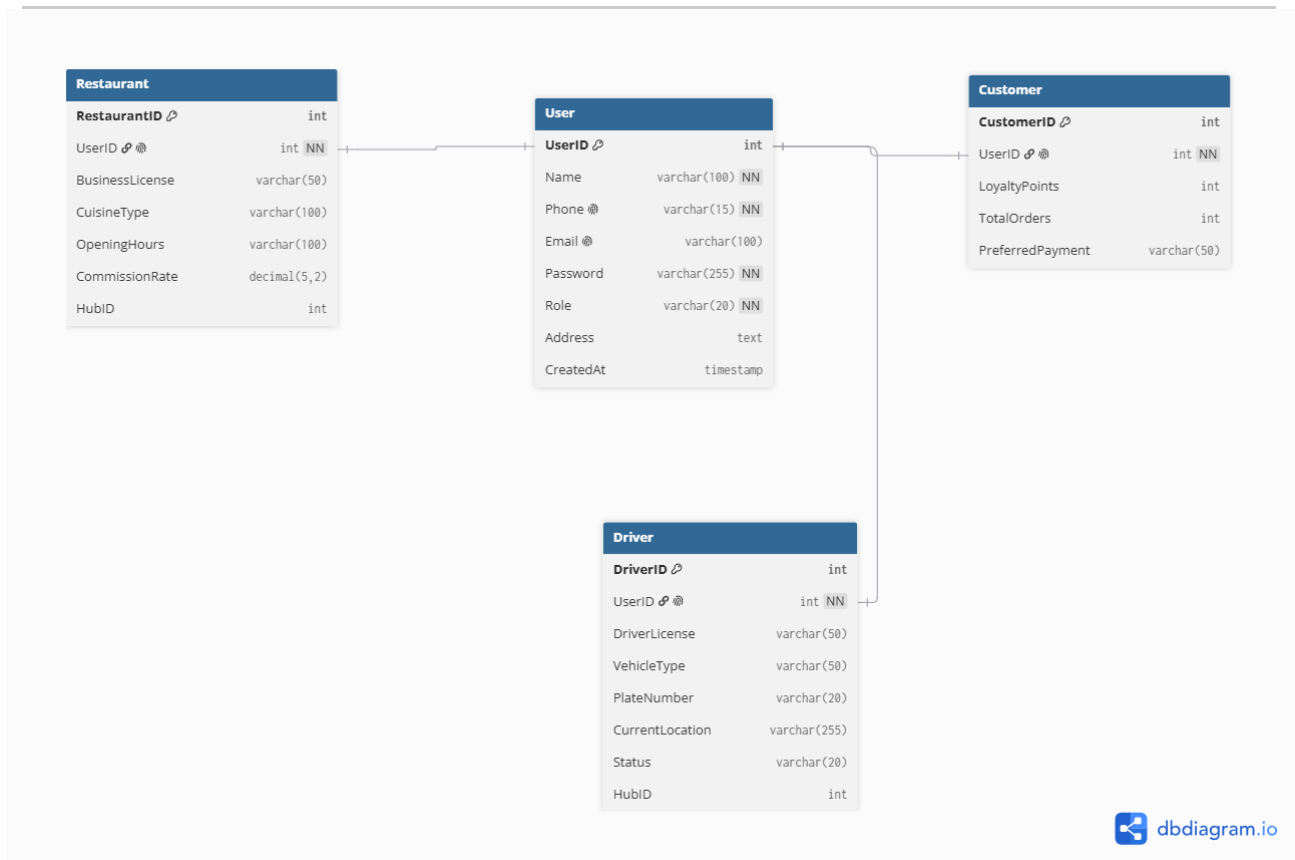
- **Integrity:** All relational mappings must strictly adhere to 3NF/BCNF normalization principles to minimize data redundancy.

### 3. BUSINESS RULES

The database schema enforces the following core business rules natively through constraints and relationships:

- **BR-01 (Account Uniqueness):** A single phone number or email address can only be associated with one User account across the entire system.
- **BR-02 (Operational Shifts):** A Driver can only be assigned to a new Order if their current status is marked as 'ONLINE'.
- **BR-03 (Order Integrity):** An OrderItem must reference a valid MenuItem, and the quantity specified must be greater than zero.
- **BR-04 (Historical Retention):** Audit logs tracking status changes must never be deleted, even if the user who initiated the change is removed from the system.
- **BR-05 (Commission Structure):** The system must default to a 15% commission rate for Restaurants unless specifically overridden during contract creation.

### 4. ENTITY-RELATIONSHIP (ER) DIAGRAM & DATA DICTIONARY



#### 4.1. User Inheritance Structure (IS-A Hierarchy)

The system implements a Superclass/Subclass (Generalization/Specialization) architecture following Extended ER (EER) principles. All system actors originate from the core `User` entity to streamline authentication and access control.

- **User (Superclass):** Manages shared identification attributes (Name, Phone, Email, Password, Address, CreatedAt). Every account in the system must exist in this table regardless of its specific role.
- **Customer (Subclass):** Represents end-users placing orders, inheriting from User. It extends core attributes with loyalty points (LoyaltyPoints), total order counts (TotalOrders), and preferred payment methods (PreferredPayment).
- **Restaurant (Subclass):** Represents merchant partners, inheriting from User. It tracks business licenses, cuisine types, opening hours, commission rates (CommissionRate), and links to operational hubs.

- **Driver (Subclass):** Represents delivery partners, inheriting from User. It expands the core profile with driver licenses, vehicle types, license plates, real-time location data, operational status, and hub assignments.

4.2. Data Dictionary (ISO/IEC 11179 Standard)

Table: User (Core Superclass)

| Attribute Name | Data Type    | Key | Nullable? | Default           | Business Rules / Constraints                                 |
|----------------|--------------|-----|-----------|-------------------|--------------------------------------------------------------|
| UserID         | INT          | PK  | No        | -                 | AUTO_INCREMENT. Unique system-wide identifier.               |
| Name           | VARCHAR(100) | -   | No        | -                 | Full name of the user.                                       |
| Phone          | VARCHAR(15)  | -   | No        | -                 | UNIQUE. Required for authentication and contact.             |
| Email          | VARCHAR(100) | -   | Yes       | -                 | UNIQUE. Contact email address.                               |
| Password       | VARCHAR(255) | -   | No        | -                 | Securely hashed password string.                             |
| Role           | ENUM         | -   | No        | -                 | Allowed values: 'Customer', 'Restaurant', 'Driver', 'Admin'. |
| Address        | TEXT         | -   | Yes       | -                 | Delivery or contact address.                                 |
| CreatedAt      | TIMESTAMP    | -   | Yes       | CURRENT_TIMESTAMP | Account registration timestamp.                              |

Table: Customer (Customer Subclass)

| Attribute Name   | Data Type   | Key | Nullable? | Default | Business Rules / Constraints                          |
|------------------|-------------|-----|-----------|---------|-------------------------------------------------------|
| CustomerID       | INT         | PK  | No        | -       | AUTO_INCREMENT.                                       |
| UserID           | INT         | FK  | No        | -       | UNIQUE. References User(UserID) ON DELETE CASCADE.    |
| LoyaltyPoints    | INT         | -   | Yes       | 0       | Accumulated membership points from successful orders. |
| TotalOrders      | INT         | -   | Yes       | 0       | Cumulative count of completed orders.                 |
| PreferredPayment | VARCHAR(50) | -   | Yes       | -       | Preferred checkout method.                            |

Table: Restaurant (Restaurant Subclass)

| Attribute Name  | Data Type    | Key | Nullable? | Default | Business Rules / Constraints                       |
|-----------------|--------------|-----|-----------|---------|----------------------------------------------------|
| RestaurantID    | INT          | PK  | No        | -       | AUTO_INCREMENT.                                    |
| UserID          | INT          | FK  | No        | -       | UNIQUE. References User(UserID) ON DELETE CASCADE. |
| BusinessLicense | VARCHAR(50)  | -   | Yes       | -       | Official business permit code.                     |
| CuisineType     | VARCHAR(100) | -   | Yes       | -       | Primary food classification.                       |
| OpeningHours    | VARCHAR(100) | -   | Yes       | -       | Standard operating hours.                          |
| CommissionRate  | DECIMAL(5,2) | -   | Yes       | 15.00   | Default platform commission fee percentage.        |
| HubID           | INT          | FK  | Yes       | NULL    | References Hub(HubID) ON DELETE SET NULL.          |

**Table: Driver (Driver Subclass)**

| Attribute Name  | Data Type    | Key | Nullable? | Default   | Business Rules / Constraints                       |
|-----------------|--------------|-----|-----------|-----------|----------------------------------------------------|
| DriverID        | INT          | PK  | No        | -         | AUTO_INCREMENT.                                    |
| UserID          | INT          | FK  | No        | -         | UNIQUE. References User(UserID) ON DELETE CASCADE. |
| DriverLicense   | VARCHAR(50)  | -   | Yes       | -         | Official driving license identification.           |
| VehicleType     | VARCHAR(50)  | -   | Yes       | -         | Mode of transport (e.g., Motorbike).               |
| PlateNumber     | VARCHAR(20)  | -   | Yes       | -         | Vehicle registration plate number.                 |
| CurrentLocation | VARCHAR(255) | -   | Yes       | -         | Real-time coordinate/location tracker.             |
| Status          | ENUM         | -   | Yes       | 'OFFLINE' | Operational state: 'ONLINE', 'OFFLINE', 'BUSY'.    |
| HubID           | INT          | FK  | Yes       | NULL      | References Hub(HubID) ON DELETE SET NULL.          |

## 5. PHASE 1 NEXT STEPS

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Moving forward into Weeks 3-4, the team will transition this problem statement and requirement set into a formalized Extended Entity-Relationship (EER) diagram. The modeling will focus on capturing the IS-A hierarchy for the User entity and accurately diagramming the 1:N and M:N relationships central to the delivery dispatch logic.